

LogTriage Project Report

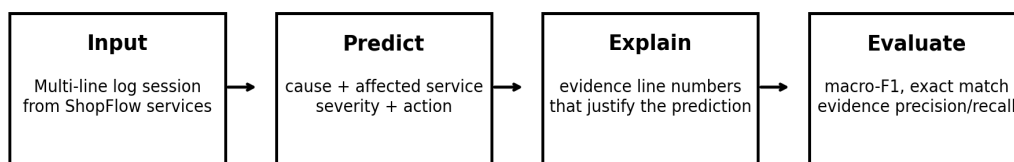
Explainable Multi-Output Triage of E-Commerce Microservice Logs

This report documents the complete course-project pipeline: task definition, synthetic data generation, dataset validation, EDA, model benchmarking, metrics, visuals, limitations, and repository artifacts.

1. Motivation and use case

In an e-commerce microservice system, checkout failures generate logs across gateway, checkout, payment, order, inventory, database, and message-queue services. Engineers need to quickly identify the failure cause, affected service, severity, recommended action, and supporting evidence lines.

LogTriage: Explainable Triage for E-commerce Microservice Logs



Novelty: not only anomaly detection - multi-output operator triage with explicit evidence lines and fixed action taxonomy.

2. ML task definition

Input: a multi-line microservice log session. Output: structured triage JSON with fixed labels and evidence-line numbers.

```
[1] gateway INFO request_id=53eefe85 POST /checkout user_id=3234
[2] cart-service INFO cart loaded cart_id=82870 item_count=7
[3] auth-service WARN token validation failed reason=expired_token user_id=1006
[4] gateway ERROR status=401 route=/checkout auth_error=invalid_signature request_id=c17ce77b
[5] checkout-service ERROR cannot continue checkout: missing valid identity context user_id=3322
```

Labels:

```
{
  "failure_cause": "authentication_failure",
  "affected_service": "auth-service",
  "severity": "medium",
  "recommended_action": "rotate_or_check_credentials",
  "evidence_lines": [
    3,
    4,
    5
  ]
}
```

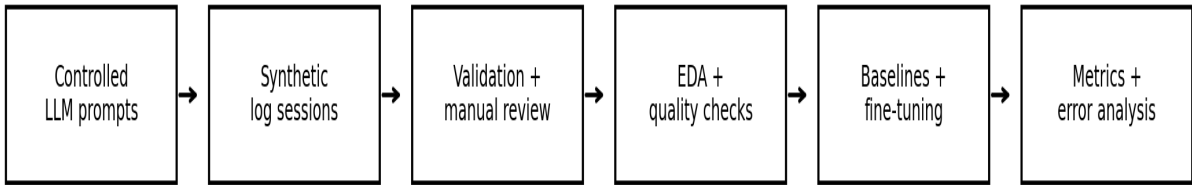
3. Failure taxonomy

Label type	Count	Examples
Failure causes	12	normal_no_issue, authentication_failure, payment_gateway_timeout, database_connection_pool_exha

Label type	Count	Examples
Affected services	7	auth-service, checkout-service, database, gateway, inventory-service, message-queue, payment-service
Severity levels	4	low, medium, high, critical
Recommended actions	12	no_action_monitor, retry_request, restart_service, rollback_deployment, scale_resource, rotate_or_che

4. Synthetic data generation method

The dataset is generated from controlled attributes: failure cause, affected service, severity, recommended action, and noise level. Each generated session has numbered log lines and explicit evidence lines. LLM prompt templates are stored in the prompts/ folder for generation and validation.

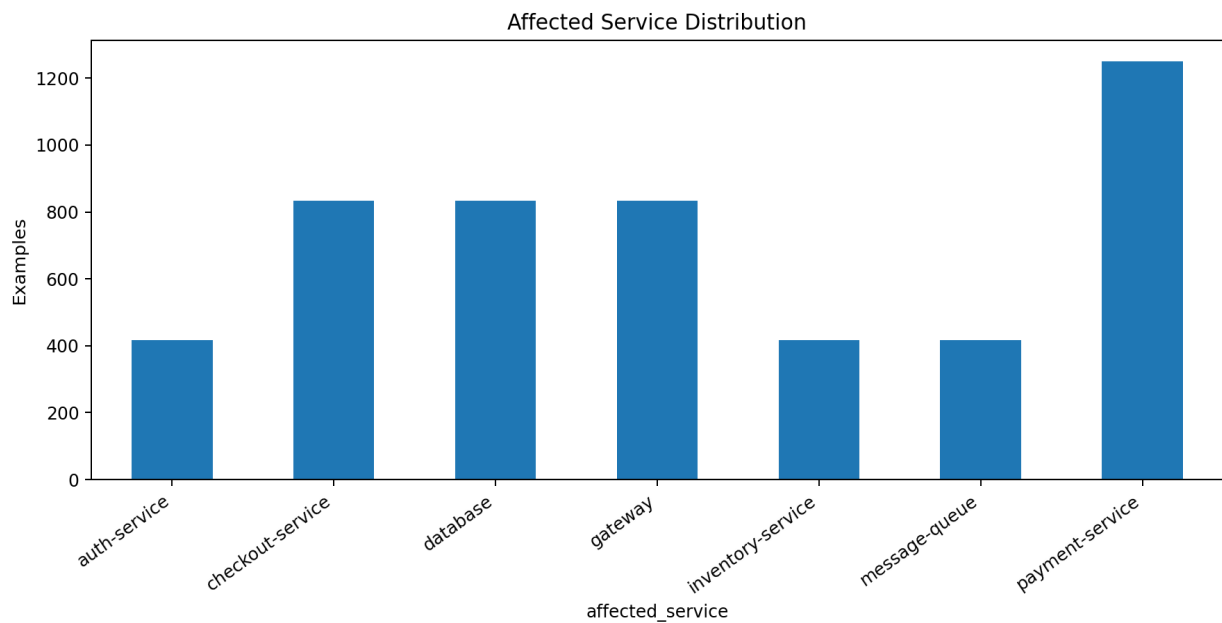
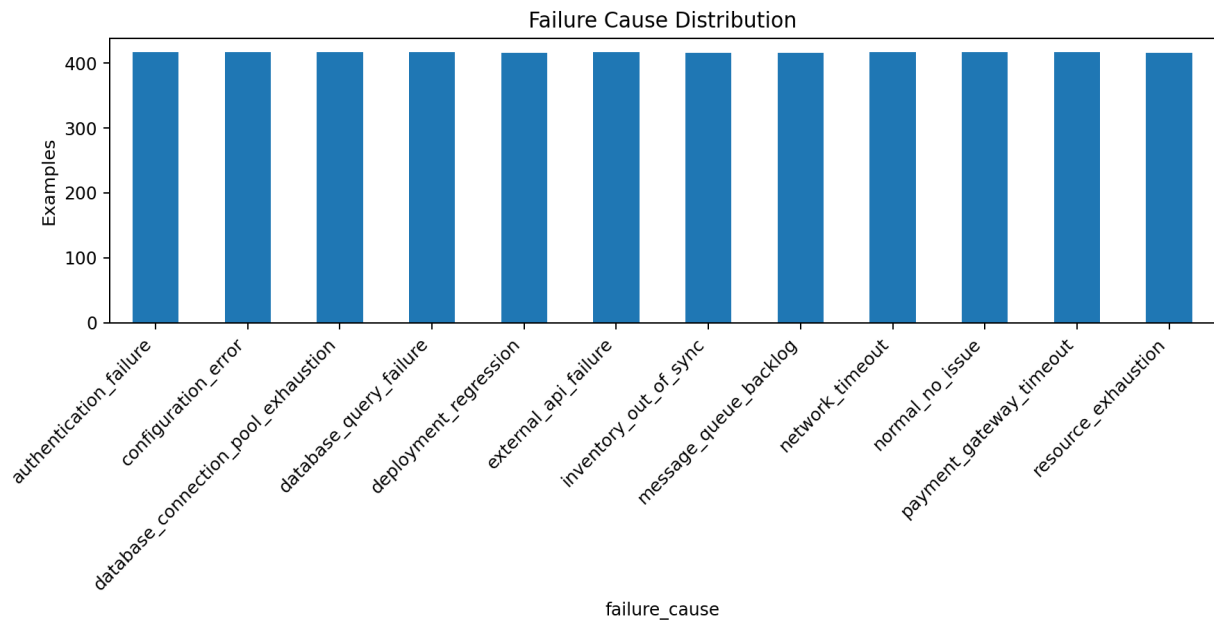


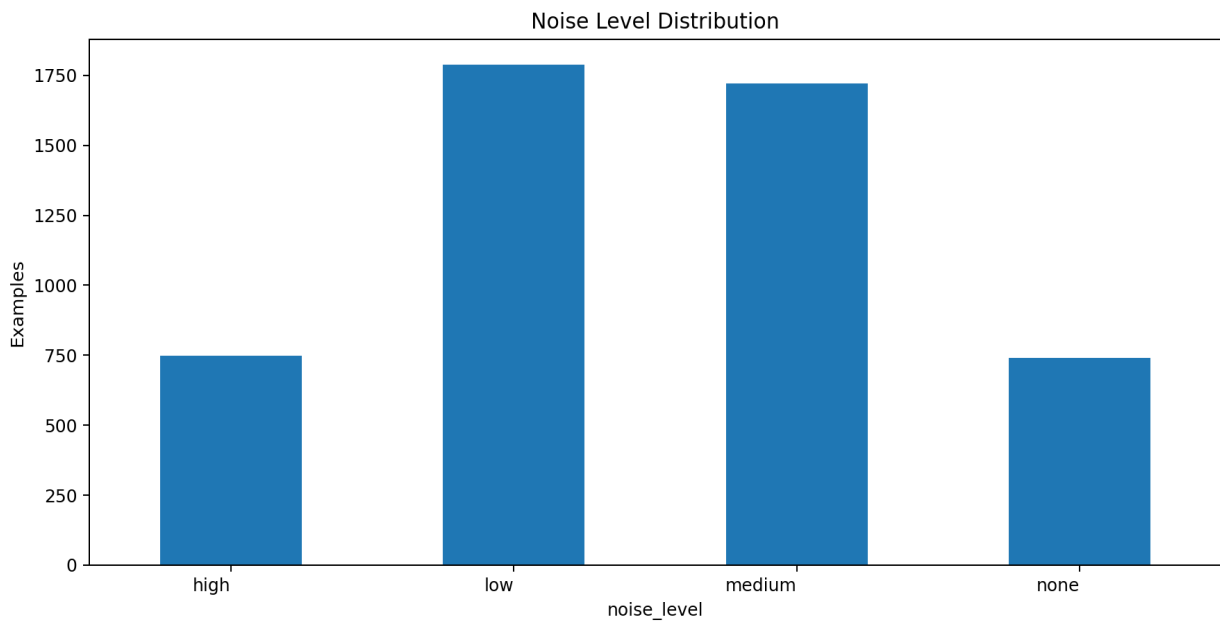
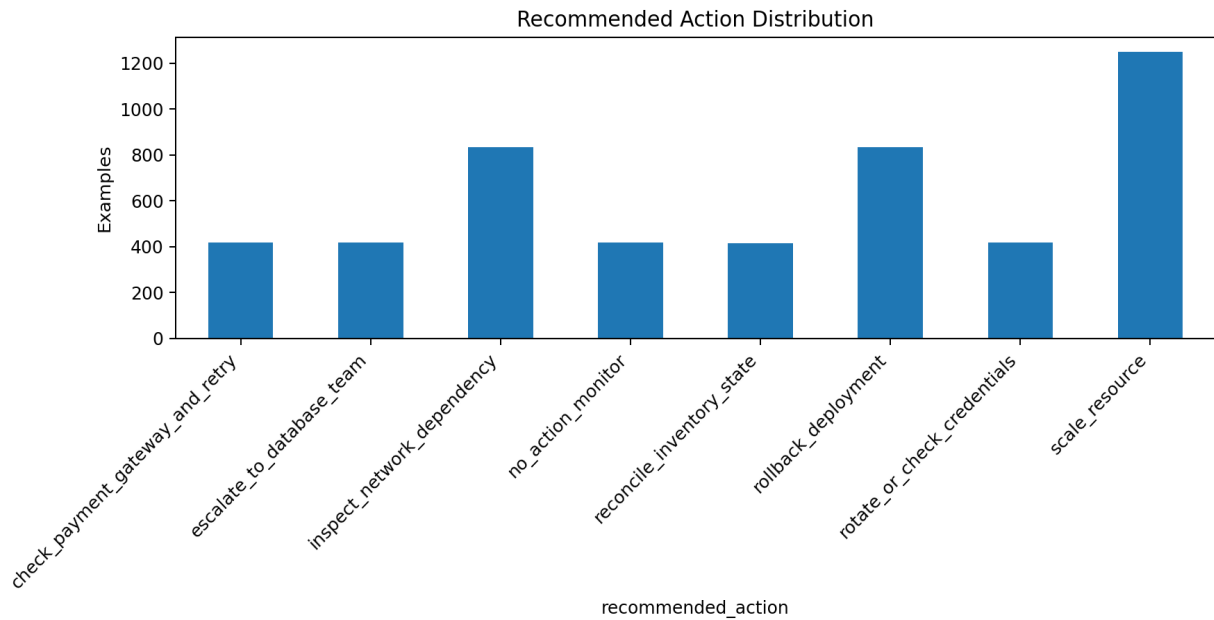
5. Dataset description

Statistic	Value
total_examples	5000.0
train_examples	3500.0
validation_examples	750.0
test_examples	750.0
failure_causes	12.0
affected_services	7.0
recommended_actions	12.0
avg_log_lines	7.84
avg_evidence_lines	2.89
validation_issues	0.0

6. Dataset EDA and quality checks

The figures below summarize class balance, severity balance, action coverage, and evidence/log-length characteristics.





7. Models and baselines

Model	Purpose	Status
Rule-based baseline	Transparent keyword/runbook baseline	Implemented and evaluated
TF-IDF + Linear SVM	Classical NLP baseline	Implemented and evaluated
Zero-shot LLM	Off-the-shelf LLM JSON prediction	Prompt/script included; API key required
Few-shot LLM	Prompt with 3-5 examples	Prompt/script included; API key required
DistilBERT/RoBERTa	Fine-tuned transformer extension	Training script included as next experiment

8. Evaluation metrics

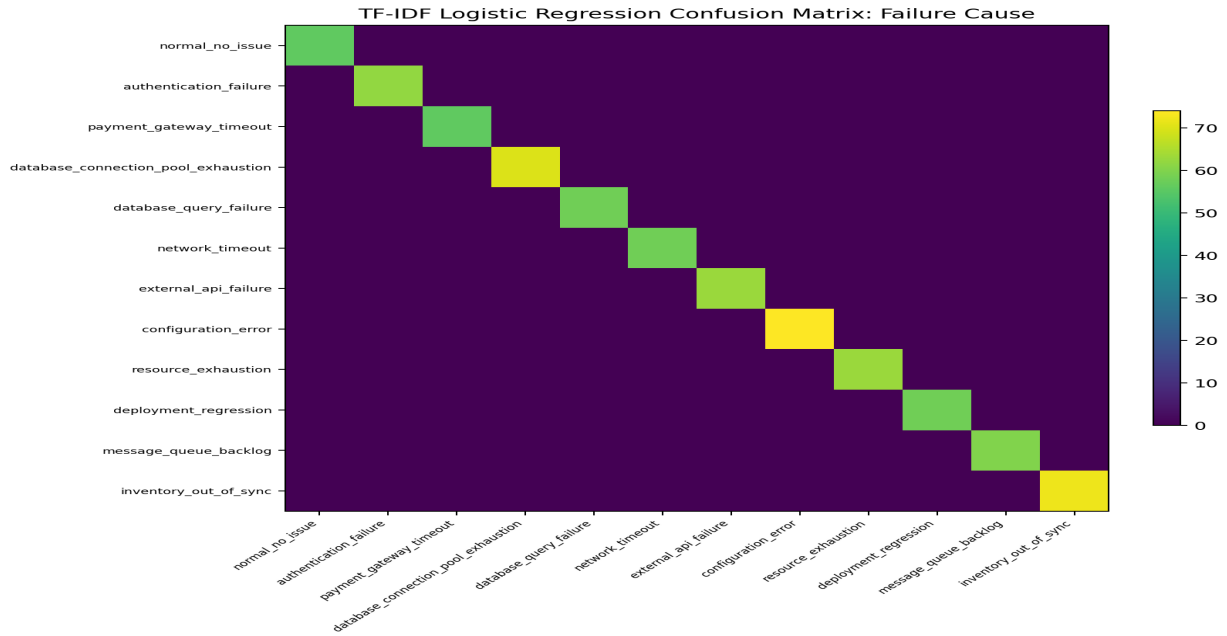
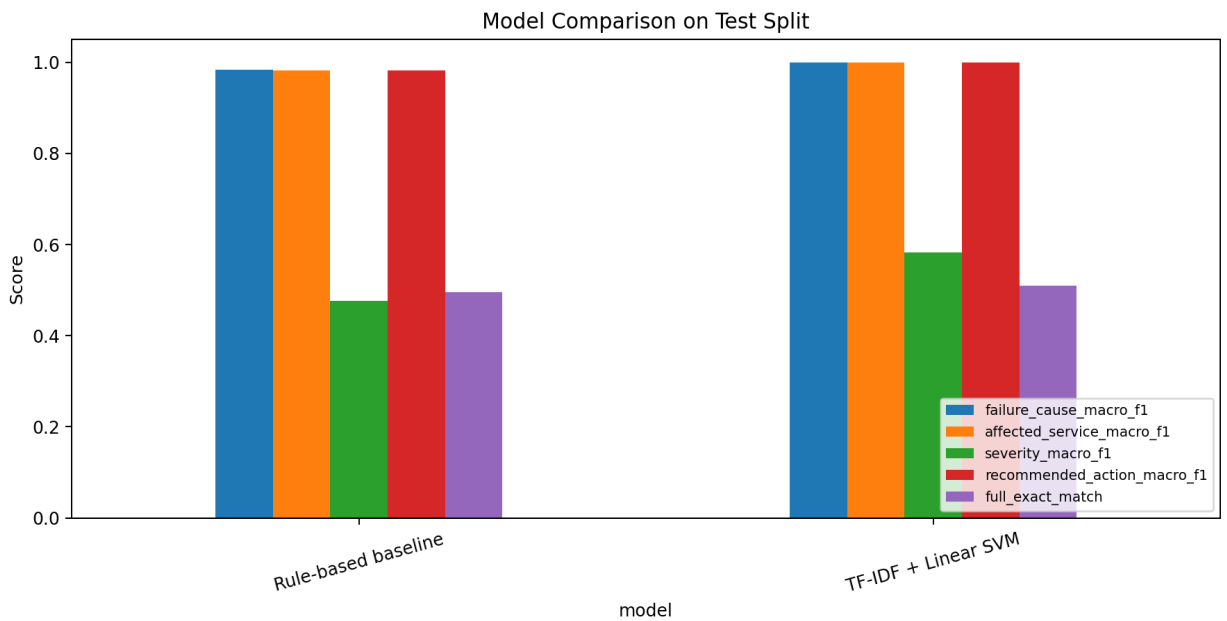
Output	Metrics
failure_cause	accuracy, macro-F1, confusion matrix
affected_service	accuracy, macro-F1
severity	accuracy, macro-F1

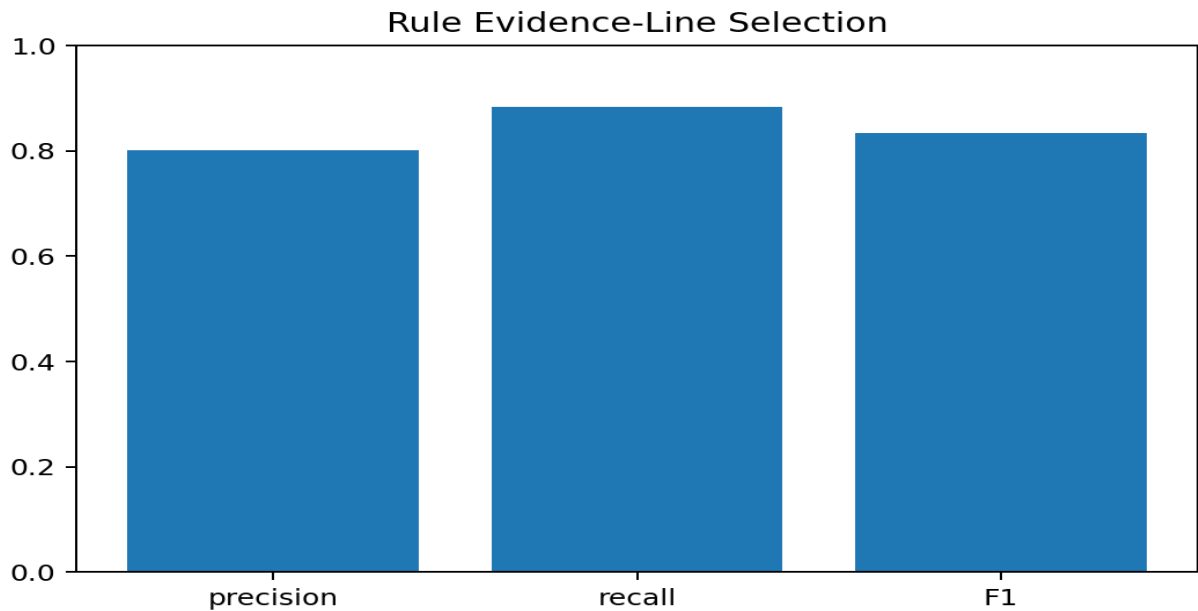
Output	Metrics
recommended_action	accuracy, macro-F1
evidence_lines	precision, recall, F1
full JSON	all-fields exact match

9. Results and analysis

Model	Cause F1	Service F1	Severity F1	Action F1	Full exact
Rule-based baseline	0.984	0.982	0.477	0.982	0.496
TF-IDF + Linear SVM	1.000	1.000	0.583	1.000	0.509

Interpretation: the current synthetic generator makes cause/service/action relatively easy because explicit keywords are often present. Severity is harder and drives the lower full exact-match score. This is useful for the final project because it identifies where better generation, LLM baselines, and fine-tuning should focus.





10. Error analysis and limitations

Important limitations: synthetic logs may not match all real production styles; labels may be simplified; action taxonomy is fixed; evidence-line evaluation can be strict; models may learn keywords instead of causal patterns. Recommended follow-up is to manually review 100 test examples, create harder ambiguous examples, run LLM baselines, and fine-tune a transformer.

11. Repository artifacts

Folder	Contents
data/	train/validation/test JSONL and label taxonomy
src/	generation, validation, baselines, LLM, transformer, evaluation
prompts/	generation, validation, zero-shot prompts
results/	metrics, predictions, evidence scores, error examples
visuals/	EDA figures, pipeline, confusion matrix, visual abstract
slides/	proposal, interim, final, and comprehensive decks
docs/	project report and previous-work summary